

Cloud Computing (Lab 11)

Name : Tooba Shafique

Reg no : 2023-BSE-065

BSE-V-B

Lab 11 – GH CLI Codespaces + AWS + Terraform: Variables, Collections, Sensitivity & EC2 Provisioning

Estimated Duration: 3 hours

Submission: Push all files and screenshots to CC_<YourName>_<YourRollNumber>/Lab11.

Task 0 – Lab Setup (Codespace & GH CLI)

Objective: Create repository, set up Codespace, and connect via SSH.

Screenshots Required:

- taskA_codespace_create_and_list.png – repo creation and Codespace list

NAME	DISPLAY NAME	REPOSITORY	BRANCH	STATE	CREATED AT
fuzzy-bassoon-4jp7p65jj46p37pwv	fuzzy bassoon	toobashafique065/Lab9	main*	Shutdown	about 5 days ago
potential-halibut-4jp7p65jqj9...	potential halibut	toobashafique065/Lab10	main*	Shutdown	about 4 days ago
automatic-fortnight-pj5r594j7...	automatic fortnight	toobashafique065/Lab11	main	Available	less than a minute ago

- taskA_codespace_ssh_connected.png – terminal inside Codespace after SSH

```
@toobashafique065 /workspaces/Lab11 (main) $ exit
logout
Connection to localhost closed.

C:\Users\malik\Lab11>gh codespace ssh -c automatic-fortnight-pj5r594j77rxfrgpx
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro
Last login: Thu Jan  1 19:26:00 2026 from ::1
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro
Last login: Thu Jan  1 19:26:00 2026 from ::1
```

Task 1 – Provider & Basic Variable (Variable Precedence)

Objective: Initialize Terraform, define variables and outputs, and test variable precedence (default, environment variable, terraform.tfvars, -var).

Screenshots Required:

- task1_touch_main_tf.png

```
@toobashafique065 @ /workspaces/Lab11 (main) $ touch main.tf
@toobashafique065 @ /workspaces/Lab11 (main) $ nano main.tf
```

- task1_main_tf_provider.png

```
GNU nano 7.2
provider "aws" {
  shared_config_files    = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}
```

- task1_terraform_init.png

```
@toobashafique065 @ /workspaces/Lab11 (main) $ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.27.0...
- Installed hashicorp/aws v6.27.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
@toobashafique065 @ /workspaces/Lab11 (main) $
```

- task1_variable_and_output_added.png

```
provider "aws" {
  shared_config_files    = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}
variable "subnet_cidr_block" {
  type = string
}

output "subnet_cidr_block_output" {
  value = var.subnet_cidr_block
}
```

- task1_apply_prompt_for_var.png

```
@toobashafique065 @ /workspaces/Lab11 (main) $ @toobashafique065 @ /workspaces/Lab11 (main) $ te
ove
var.subnet_cidr_block
Enter a value: yes

Changes to Outputs:
  + subnet_cidr_block_output = "yes"

You can apply this plan to save these new output values to the Terraform state, without changing
infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
subnet_cidr_block_output = "yes"
```

- task1_apply_with_default.png

```

Outputs:
subnet_cidr_block_output = "yes"
@toobashafique065 /workspaces/Lab11 (main) $ nano main.tf
@toobashafique065 /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  ~ subnet_cidr_block_output = "yes" -> "10.0.0.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real
infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

```

- task1_env_var_set_and_apply.png

```

@toobashafique065 /workspaces/Lab11 (main) $ export TF_VAR_subnet_cidr_block=10.0.20.0/24
@toobashafique065 /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  ~ subnet_cidr_block_output = "10.0.0.0/24" -> "10.0.20.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real
infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
subnet_cidr_block_output = "10.0.20.0/24"
@toobashafique065 /workspaces/Lab11 (main) $

```

- task1_terraform_tfvars_and_apply.png

```

@toobashafique065 /workspaces/Lab11 (main) $ vim terraform.tfvars
@toobashafique065 /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  ~ subnet_cidr_block_output = "10.0.20.0/24" -> "10.0.30.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real
infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
subnet_cidr_block_output = "10.0.30.0/24"
@toobashafique065 /workspaces/Lab11 (main) $

```

- task1_var_override_with_dash_var.png

```

@toobashafique065 /workspaces/Lab11 (main) $ terraform apply -auto-approve -var "subnet_cidr_block=10.0.40.0/24"

Changes to Outputs:
  ~ subnet_cidr_block_output = "10.0.30.0/24" -> "10.0.40.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real
infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
subnet_cidr_block_output = "10.0.40.0/24"

```

- task1_printenv_tf_var_and_unset.png

```

@toobashafique065 /workspaces/Lab11 (main) $ printenv | grep TF_VAR_
TF_VAR_subnet_cidr_block=10.0.20.0/24
@toobashafique065 /workspaces/Lab11 (main) $ unset TF_VAR_subnet_cidr_block
@toobashafique065 /workspaces/Lab11 (main) $ printenv | grep TF_VAR_
@toobashafique065 /workspaces/Lab11 (main) $

```

Task 2 – Variable Validation & Sensitive / Ephemeral Variables

Objective: Use variable validation, create sensitive and ephemeral variables, and observe behavior in Terraform state and outputs.

Screenshots Required:

- task2_subnet_variable_with_validation.png

```

es Command Prompt - gni codespace ssh -t automatic-terraform-jps594j77xngpx
GNU nano 7.2 main.tf *
provider "aws" {
  shared_config_files = ["~/aws/config"]
  shared_credentials_files = ["~/aws/credentials"]
}
variable "subnet_cidr_block" {
  type = string
  default = ""
  description = "CIDR block to assign to the application subnet"
  sensitive = false
  nullable = false
  ephemeral = false

  validation {
    condition = can(regex("^[0-9]{1,3}\\.[0-9]{1,3}\\.[0-9]{1,3}/[0-9]+$", var.subnet_cidr_block))
    error_message = "The subnet_cidr_block must be a valid CIDR notation string, such as 10.0.0.0/24."
  }
}

```

- task2_subnet_validation_error.png

```

st00bashafique065 @ /workspaces/Lab11 (main) $ terraform apply -auto-approve -var "subnet_cidr_block=10.0.0"
Error: Invalid value for variable

on main.tf line 5:
5: variable "subnet_cidr_block" {
  var.subnet_cidr_block is "10.0.0"

The subnet_cidr_block must be a valid CIDR notation string, such as 10.0.0.0/24.
This was checked by the validation rule at main.tf:13,3-13.

```

- task2_api_token_variable_added.png

```

variable "api_session_token" {
  type = string
  default = ""
  description = "Short-lived API session token used during apply operations"
  sensitive = true
  nullable = false
  ephemeral = false

  validation {
    condition = can(regex("[A-Za-z0-9-]{20,}", var.api_session_token))
    error_message = "The API session token must be at least 20 characters and contain only letters, number"
  }
}

output "api_session_token_output" {
  value = var.api_session_token
  sensitive = true
}

```

- task2_api_token_apply_sensitive.png

```

st00bashafique065 @ /workspaces/Lab11 (main) $ terraform apply -auto-approve -var "subnet_cidr_block=10.0.0.0/24" -var
api_session_token=my_API_session_Token_1234567890"
Changes to Outputs:
+ api_session_token_output = (sensitive value)
~ subnet_cidr_block_output = "10.0.40.0/24" -> "10.0.0.0/24"
You can apply this plan to save these new output values to the Terraform state, without changing any real
infrastructure.
Apply complete! Resources: 0 added, 0 changed, 0 destroyed.
Outputs:
api_session_token_output = <sensitive>
subnet_cidr_block_output = "10.0.0.0/24"

```

- task2_check_terraform_state_api_token.png

```
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 6,
  "lineage": "cid66ff4-1548-5719-54a5-afb90a6ec625",
  "outputs": {
    "api_session_token_output": {
      "value": "my_API_session_Token_1234567890",
      "type": "string",
      "sensitive": true
    },
    "subnet_cidr_block_output": {
      "value": "10.0.0.0/24",
      "type": "string"
    }
  },
  "resources": [],
  "check_results": [
    {
      "object_kind": "var",
      "config_addr": "var.subnet_cidr_block",
      "status": "pass",
      "objects": [
        {
          "object_addr": "var.subnet_cidr_block",
          "status": "pass"
        }
      ]
    },
    {
      "object_kind": "var",
      "config_addr": "var.api_session_token",
      "status": "pass",
      "objects": [
        {
          "object_addr": "var.api_session_token",
          "status": "pass"
        }
      ]
    }
  ]
}
```

- task2_api_token_ephemeral_error.png

```
@toobashafique065 @ /workspaces/Lab11 (main) $ nano main.tf
@toobashafique065 @ /workspaces/Lab11 (main) $ terraform apply -auto-approve -var "subnet_cidr_block=10.0
api_session_token=my_API_session_Token_1234567890"

Error: Ephemeral value not allowed

on main.tf line 38, in output "api_session_token_output":
38:   value      = var.api_session_token

This output value is not declared as returning an ephemeral value, so it cannot be set to a result deriv
ephemeral value.
```

- task2_api_token_default_apply.png

```
@toobashafique065 @ /workspaces/Lab11 (main) $ nano main.tf
@toobashafique065 @ /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  ~ subnet_cidr_block_output = "10.0.0.0/24" -> "10.0.30.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real
infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = <sensitive>
subnet_cidr_block_output = "10.0.30.0/24"
@toobashafique065 @ /workspaces/Lab11 (main) $
```

Task 3 – Project-level Variables, Locals, and Outputs

Objective: Define project-level variables, create locals for derived values, and configure outputs to display them.

Screenshots Required:

- task3_variables_added.png

```
variable "environment" {}
variable "project_name" {}
variable "primary_subnet_id" {}
variable "subnet_count" {}
variable "monitoring" {}
```

- task3_terraform_tfvars_populated.png

```

subnets_cidr_block = "10.0.30.0/24"
environment         = "dev"
project_name        = "lab_work"
primary_subnet_id   = "subnet-0464b8ab5b6f1ce44"
subnet_count        = 3
monitoring           = true

```

- task3_locals_tf_created.png

```

GNU nano 7.2                                locals
locals {
  resource_name = "${var.project_name}-${var.environment}"
  primary_public_subnet = var.primary_subnet_id
  subnet_count        = var.subnet_count
  is_production        = var.environment == "prod"
  monitoring_enabled   = var.monitoring || local.is_production
}

```

- task3_outputs_apply.png

```

root@asharique05: /workspaces/Lab11 (main) $ terraform apply -auto-approve
Changes to Outputs:
  + is_production          = false
  + monitoring_enabled     = true
  + primary_public_subnet = "subnet-0464b8ab5b6f1ce44"
  + resource_name         = "lab_work-dev"
  + subnet_count          = 3

You can apply this plan to save these new output values to the Terraform state, without changing any real
infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = <sensitive>
is_production            = false
monitoring_enabled       = true
primary_public_subnet    = "subnet-0464b8ab5b6f1ce44"
resource_name            = "lab_work-dev"
subnet_cidr_block_output = "10.0.30.0/24"
subnet_count             = 3
root@asharique05: /workspaces/Lab11 (main) $

```

Task 4 – Maps and Objects

Objective: Use map and object variables, populate them in terraform.tfvars, and output the values.

Screenshots Required:

- task4_tags_variable_added.png

```

variable "tags" {
  type = map(string)
}

output "tags" {
  value = var.tags
}

```

- task4_tags_output.png

```

root@asharique05: /workspaces/Lab11 (main) $ terraform apply -auto-approve
Changes to Outputs:
  + tags = {
    + Environment = "dev"
    + Owner       = "platform-team"
    + Project     = "sample-app"
  }

You can apply this plan to save these new output values to the Terraform state, without changing any real
infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = <sensitive>
is_production            = false
monitoring_enabled       = true
primary_public_subnet    = "subnet-0464b8ab5b6f1ce44"
resource_name            = "lab_work-dev"
subnet_cidr_block_output = "10.0.30.0/24"
subnet_count             = 3
tags = tomap({
  "Environment" = "dev"
  "Owner"       = "platform-team"
  "Project"     = "sample-app"
})
root@asharique05: /workspaces/Lab11 (main) $

```

- task4_server_config_output.png

```

stoobashafique005 @ /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  + server_config = {
    + backup_enabled = false
    + instance_type  = "t3.micro"
    + monitoring     = true
    + name           = "web-server"
    + storage_gb     = 20
  }

You can apply this plan to save these new output values to the Terraform state, without changing any real
infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = <sensitive>
is_production = false
monitoring_enabled = true
primary_public_subnet = "subnet-0464b8ab5b6f1ce44"
resource_name = "lab_work-dev"
server_config = {
  "backup_enabled" = false
  "instance_type"  = "t3.micro"
  "monitoring"     = true
  "name"           = "web-server"
  "storage_gb"     = 20
}
subnet_cidr_block_output = "10.0.30.0/24"
subnet_count = 3
tags = tomap({
  "Environment" = "dev"
  "Owner"       = "platform-team"
  "Project"     = "sample-app"
})

```

Task 5 – Collections: List, Tuple, Set & Mutation via Locals

Objective: Define list, tuple, and set variables, observe behavior, mutate collections via locals, and compare original vs mutated values.

Screenshots Required:

- task5_collections_defined.png

```

output "server_config" {
  value = var.server_config
}

variable "server_names" {
  type     = list(string)
  default = ["web-2", "web-1", "web-2"]
}

variable "server_metadata" {
  type     = tuple([string, number, bool])
  default = ["web-1", 4, true]
}

variable "availability_zones" {
  type     = set(string)
  default = ["me-central-1b", "me-central-1a", "me-central-1b"]
}

output "compare_collections" {
  value = {
    list_example  = var.server_names
    tuple_example = var.server_metadata
    set_example   = var.availability_zones
  }
}

```

- task5_compare_collections.png

Outputs:

```
api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-2",
    "web-1",
    "web-2",
  ])
  "set_example" = toset([
    "me-central-1a",
    "me-central-1b",
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
is_production = false
monitoring_enabled = true
primary_public_subnet = "subnet-0464b8ab5b6f1ce44"
resource_name = "lab_work-dev"
server_config = {
  "backup_enabled" = false
  "instance_type" = "t3.micro"
  "monitoring" = true
  "name" = "web-server"
  "storage_gb" = 20
}
subnet_cidr_block_output = "10.0.30.0/24"
subnet_count = 3
tags = tomap({
  "Environment" = "dev"
  "Owner" = "platform-team"
  "Project" = "sample-app"
}))
```

- task5_locals_mutations.png

```
locals {
  mutated_list  = setunion(var.server_names, ["web-3"])
  mutated_tuple = setunion(var.server_metadata, ["web-2"])
  mutated_set   = setunion(var.availability_zones, ["me-central-1c"])
}
```

- task5_mutation_comparison.png


```

@toobashafique065 @ /workspaces/Lab11 (main) $ nano main.tf
@toobashafique065 @ /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
+ mutation_comparison = {
+   mutated_tuple = [
+     "4",
+     "true",
+     "web-1",
+     "web-2",
+   ]
+   original_tuple = [
+     "web-1",
+     4,
+     true,
+   ]
+ }

You can apply this plan to save these new output values to the Terraform state, without
infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-2",
    "web-1",
    "web-2",
  ])
  "set_example" = toset([
    "me-central-1a",
    "me-central-1b",
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
is_production = false
monitoring_enabled = true

```

Task 6 – Null, Any Type & Dynamic Values

Objective: Work with null values, dynamic (any) types, and optional tags.

Screenshots Required:

- task6_optional_tag_variable.png

```

variable "optional_tag" {
  type        = string
  description = "A tag that may or may not be provided"
  default     = null
}

```

- task6_locals_merge.png

```

}
locals {
  server_tags = merge(
    { Name = "web-server" },
    var.optional_tag != null ? { Custom = var.optional_tag } : {}
  )
}

```

- task6_optional_tag_no_value.png

```
@toobashafique065 @ /workspaces/Lab11 (main) $ nano main.tf
@toobashafique065 @ /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  + optional_tag              = {
    + Name = "web-server"
  }

You can apply this plan to save these new output values to the Terraform state, without changing any real
infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-2",
    "web-1",
    "web-2",
  ])
  "set_example" = toset([
    "me-central-1a",
    "me-central-1b",
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
```

- task6_optional_tag_with_value.png

```
@toobashafique065 @ /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  ~ optional_tag              = {
    + Custom = "dev"
    # (1 unchanged attribute hidden)
  }

You can apply this plan to save these new output values to the Terraform state, without changing any real
infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-2",
    "web-1",
    "web-2",
  ])
  "set_example" = toset([
    "me-central-1a",
    "me-central-1b",
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
```

- task6_dynamic_value_string.png

```
dynamic_value = "hello"
```

```
@toobashafique065 @ /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
  + value_received            = "hello"

You can apply this plan to save these new output values to the Terraform state, without changing any real
infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-2",
  ])
}
```

- task6_dynamic_value_number.png

```
@toobashafique065 [ /workspaces/Lab11 (main) ] $ nano terraform.tfvars
@toobashafique065 [ /workspaces/Lab11 (main) ] $ terraform apply -auto-approve

Changes to Outputs:
  ~ value_received      = "hello" -> "42"

You can apply this plan to save these new output values to the Terraform state, without changing any real
infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-2",
    "web-1",
  ])
```

- task6_dynamic_value_list.png

```
@toobashafique065 [ /workspaces/Lab11 (main) ] $ terraform apply -auto-approve

Changes to Outputs:
  ~ value_received      = "42" -> [
    + "a",
    + "b",
    + "c",
  ]

You can apply this plan to save these new output values to the Terraform state, without changing a
infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-2",
    "web-1",
    "web-2",
  ])
  "set_example" = toset([
    "me-central-1a",
    "me-central-1b",
  ])
```

- task6_dynamic_value_map.png

```
@toobashafique065 [ /workspaces/Lab11 (main) ] $ terraform apply -auto-approve

Changes to Outputs:
  ~ value_received      = [
    - "a",
    - "b",
    - "c",
  ] -> {
    + cpu = 4
    + name = "server"
  }

You can apply this plan to save these new output values to the Terraform state, without changing
infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
```

- task6_dynamic_value_null.png

```

@toobashafique065 /workspaces/Lab11 (main) $ terraform apply -auto-approve

Changes to Outputs:
- value_received = {
  - cpu = 4
  - name = "server"
} -> null

You can apply this plan to save these new output values to the Terraform state, without changing
infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-2",
    "web-1",
    "web-2",
  ])
  "set_example" = toset([
    "me-central-1a",
    "me-central-1b",
  ])
  "tuple_example" = [

```

Task 7 – Git Ignore

Objective: Create .gitignore to prevent committing sensitive files like .tfstate, .pem, and credentials.

Screenshots Required:

- task7_gitignore_created.png

```

@toobashafique065 /workspaces/Lab11 (main) $ touch .gitignore
@toobashafique065 /workspaces/Lab11 (main) $ nano .gitignore
@toobashafique065 /workspaces/Lab11 (main) $ cat .gitignore
.terraform/*
*.tfstate
*.tfstate.*
*.tfvars
*.pem

```

Task 8 – Build Real Infrastructure: VPC, Subnet, IGW, Routing

Objective: Create AWS VPC, Subnet, Internet Gateway, Route Tables (custom and default), and associate them properly.

Screenshots Required:

- task8_clean_files.png

```

GNU nano 7.2
provider "aws" {
  shared_config_files = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

```

- task8_variables_recreated.png

```

variable "vpc_cidr_block" {}
variable "subnet_cidr_block" {}
variable "availability_zone" {}
variable "env_prefix" {}

```

- task8_vpc_resources_added.png

```

resource "aws_vpc" "myapp_vpc" {
  cidr_block = var.vpc_cidr_block
  tags = {
    Name = "${var.env_prefix}-vpc"
  }
}

```

- task8_subnet_resources_added.png

```

resource "aws_subnet" "myapp_subnet_1" {
  vpc_id            = aws_vpc.myapp_vpc.id
  cidr_block        = var.subnet_cidr_block
  availability_zone  = var.availability_zone
  tags = {
    Name = "${var.env_prefix}-subnet-1"
  }
}

```

- task8_terraform_tfvars_vpc_values.png

```

vpc_cidr_block      = "10.0.0.0/16"
subnet_cidr_block   = "10.0.10.0/24"
availability_zone    = "me-central-1a"
env_prefix          = "dev"

```

- task8_vpc_subnet_apply.png

```

$ cd ~/workspace/lab11 (main) $ terraform apply -auto-approve
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
  + create

Terraform will perform the following actions:

  # aws_subnet.myapp_subnet_1 will be created
  + resource "aws_subnet" "myapp_subnet_1" {
    + arn                                     = (known after apply)
    + assign_ipv6_address_on_creation         = false
    + availability_zone                       = "me-central-1a"
    + availability_zone_id                   = (known after apply)
    + cidr_block                             = "10.0.10.0/24"
    + enable_dns64                           = false
    + enable_resource_name_dns_a_record_on_launch = false
    + enable_resource_name_dns_aaaa_record_on_launch = false
    + id                                     = (known after apply)
    + ipv6_cidr_block_association_id         = (known after apply)
    + ipv6_native                             = false
    + map_public_ip_on_launch                = false
    + owner_id                              = (known after apply)
    + private_dns_hostname_type_on_launch    = (known after apply)
    + region                                 = "me-central-1"
    + tags                                  = {
      + "Name" = "dev-subnet-1"
    }
    + tags_all                              = {
      + "Name" = "dev-subnet-1"
    }
  }

```

```

- tags
  - Environment = "dev" = {
    - Owner      = "platform-team"
    - Project    = "sample-app"
  }
} -> null

aws_vpc.myapp_vpc: Creating...
aws_vpc.myapp_vpc: Creation complete after 1s [id=vpc-04dabca6095a83798]
aws_subnet.myapp_subnet_1: Creating...
aws_subnet.myapp_subnet_1: Creation complete after 1s [id=subnet-07bbc7d18a3245a8f]

```

- task8_igw_route_table_before_apply.png

```

resource "aws_vpc" "myapp_vpc" {
  cidr_block = var.vpc_cidr_block
  tags = {
    Name = "${var.env_prefix}-vpc"
  }
}
resource "aws_subnet" "myapp_subnet_1" {
  vpc_id = aws_vpc.myapp_vpc.id
  cidr_block = var.subnet_cidr_block
  availability_zone = var.availability_zone
  tags = {
    Name = "${var.env_prefix}-subnet-1"
  }
}
resource "aws_internet_gateway" "myapp_igw" {
  vpc_id = aws_vpc.myapp_vpc.id
  tags = {
    Name = "${var.env_prefix}-igw"
  }
}
resource "aws_route_table" "myapp_route_table" {
  vpc_id = aws_vpc.myapp_vpc.id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.myapp_igw.id
  }

  tags = {
    Name = "${var.env_prefix}-rt"
  }
}

```

- task8_igw_route_table_after_apply.png

```

+ tags = {
+   "Name" = "dev-igw"
+ }
+ tags_all = {
+   "Name" = "dev-igw"
+ }
+ vpc_id = "vpc-04dabca6095a83798"
+ }

# aws_route_table.myapp_route_table will be created
+ resource "aws_route_table" "myapp_route_table" {
+   arn = (known after apply)
+   id = (known after apply)
+   owner_id = (known after apply)
+   propagating_vgws = (known after apply)
+   region = "me-central-1"
+   route = [
+     {
+       cidr_block = "0.0.0.0/0"
+       gateway_id = (known after apply)
+     }
+   ]
+   tags = {
+     "Name" = "dev-rt"
+   }
+   tags_all = {
+     "Name" = "dev-rt"
+   }
+   vpc_id = "vpc-04dabca6095a83798"
+ }

Plan: 2 to add, 0 to change, 0 to destroy.
aws_internet_gateway.myapp_igw: Creating...
aws_internet_gateway.myapp_igw: Creation complete after 0s [id=igw-0cc1c18b3a01e71b0]
aws_route_table.myapp_route_table: Creating...
aws_route_table.myapp_route_table: Creation complete after 1s [id=rtb-0f98fe6cafe7918c1]
Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

```

- task8_association_apply.png

```

@toobashafique065 @ /workspaces/Lab11 (main) $ nano main.tf
@toobashafique065 @ /workspaces/Lab11 (main) $ terraform apply -auto-approve
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-04dabca6095a83798]
aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-0cc1c18b3a01e71b0]
aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-07bbc7d18a3245a8f]
aws_route_table.myapp_route_table: Refreshing state... [id=rtb-0f98fe6cafe7918c1]

Terraform used the selected providers to generate the following execution plan. Resource actions
following symbols:
+ create

Terraform will perform the following actions:

# aws_route_table_association.a_rtb_subnet will be created
+ resource "aws_route_table_association" "a_rtb_subnet" {
+   id = (known after apply)
+   region = "me-central-1"
+   route_table_id = "rtb-0f98fe6cafe7918c1"
+   subnet_id = "subnet-07bbc7d18a3245a8f"
+ }

Plan: 1 to add, 0 to change, 0 to destroy.
aws_route_table_association.a_rtb_subnet: Creating...
aws_route_table_association.a_rtb_subnet: Creation complete after 1s [id=rtbassoc-0b1169b8a3a]
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

```

- task8_default_route_table.png

```
resource "aws_default_route_table" "main_rt" {
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.myapp_igw.id
  }

  tags = {
    Name = "${var.env_prefix}-rt"
  }
}
```

- task8_default_route_table_apply.png

```

} -> null
- tags_all = {
  - "Name" = "dev-rt"
} -> null
- vpc_id = "vpc-04dabca6095a83798" -> null
}

# aws_route_table_association.a_rtb_subnet will be destroyed
# (because aws_route_table_association.a_rtb_subnet is not in configuration)
- resource "aws_route_table_association" "a_rtb_subnet" {
  - id = "rtbassoc-0b1169b8a3a379e93" -> null
  - region = "me-central-1" -> null
  - route_table_id = "rtb-0f98fe6cafe7918c1" -> null
  - subnet_id = "subnet-07bbc7d18a3245a8f" -> null
  # (1 unchanged attribute hidden)
}

Plan: 1 to add, 0 to change, 2 to destroy.
aws_route_table_association.a_rtb_subnet: Destroying... [id=rtbassoc-0b1169b8a3a379e93]
aws_default_route_table.main_rt: Creating...
aws_route_table_association.a_rtb_subnet: Destruction complete after 1s
aws_route_table.myapp_route_table: Destroying... [id=rtb-0f98fe6cafe7918c1]
aws_default_route_table.main_rt: Creation complete after 1s [id=rtb-0fa00af3be972e7d0]
aws_route_table.myapp_route_table: Destruction complete after 0s

Apply complete! Resources: 1 added, 0 changed, 2 destroyed.
root@ec2-149-005 ~ # ./workspace/lab11/main.sh & exit
```

Your VPCs (2) Info Last updated less than a minute ago Actions

Find VPCs by attribute or tag

☐	Name	VPC ID	State	Encryption
☐	-	vpc-0b8fa1602193cc7f7	Available	-
☐	dev-vpc	vpc-04dabca6095a83798	Available	-

Task 9 – Security Group, Key Pair, EC2 Instance, user_data & Nginx

Objective: Create Security Group, Key Pair, EC2 instance, configure SSH, install nginx via inline and external script user_data.

Screenshots Required:

- task9_my_ip_variable_added.png

```
variable "env_prefix" {}
variable "my_ip" {}
variable "instance_type" {}
```

- task9_public_ip_curl.png

```
my_ip = "20.192.21.49/32"
instance_type = "t3.micro"
availability_zone = "me-central-1a"
env_prefix = "dev"
```

- task9_security_group_apply.png

```

$ terraform apply -auto-approve
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-04dabca6095a83798]
aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-0cc1c18b3a01e71b0]
aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-07bbc7d18a3245a8f]
aws_default_route_table.main_rt: Refreshing state... [id=rtb-0fa00af3be972e7d0]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
+ create

Terraform will perform the following actions:

# aws_default_security_group.myapp_sg will be created
+ resource "aws_default_security_group" "myapp_sg" {
+   arn              = (known after apply)
+   description      = (known after apply)
+   egress            = [
+     {
+       cidr_blocks = [
+         "0.0.0.0/0",
+       ]
+       from_port    = 0
+       ipv6_cidr_blocks = []
+       prefix_list_ids = []
+       protocol      = "tcp"
+       security_groups = []
+       self          = false
+       to_port       = 0
+     }
+   ],
+   id              = (known after apply)
+   ingress          = [
+     {
+       cidr_blocks = [
+         "0.0.0.0/0",

```

- task9_keypair_created_and_saved.png
- task9_instance_type_set.png

```

resource "aws_instance" "myapp-server" {
  ami              = "ami-05524d6658fcf35b6"
  instance_type    = var.instance_type
  subnet_id        = aws_subnet.myapp_subnet_1.id
  security_groups   = [aws_default_security_group.myapp_sg.id]
  availability_zone = var.availability_zone
  associate_public_ip_address = true
  key_name          = "MyED25519Key"

  tags = {
    Name = "${var.env_prefix}-ec2-instance"
  }
}

output "aws_instance_public_ip" {
  value = aws_instance.myapp-server.public_ip
}

```

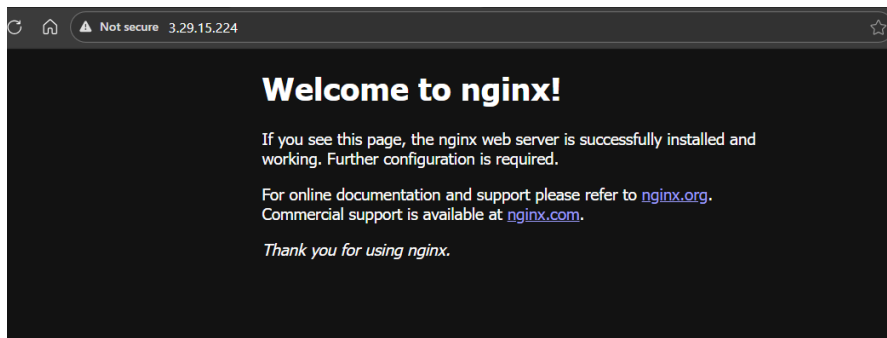
- task9_ec2_apply_and_public_ip.png


```
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

- task9_nginx_browser_page.png



Cleanup

Objective: Destroy all Terraform-managed resources, verify state files, and ensure no sensitive files are committed.

Screenshots Required:

- cleanup_destroy.png

```
Plan: 0 to add, 0 to change, 6 to destroy.

Changes to Outputs:
  - aws_instance_public_ip = "3.29.15.224" -> null
aws_default_route_table.main_rt: Destroying... [id=rtb-0fa0af3be972e7d0]
aws_default_route_table.main_rt: Destruction complete after 0s
aws_instance_myapp_server: Destroying... [id=i-095776ece185713ae]
aws_internet_gateway.myapp_igw: Destroying... [id=igw-0ccc1c18b3a01e71b0]
aws_instance_myapp_server: Still destroying... [id=i-095776ece185713ae, 00m10s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0ccc1c18b3a01e71b0, 00m10s elapsed]
aws_instance_myapp_server: Still destroying... [id=i-095776ece185713ae, 00m20s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0ccc1c18b3a01e71b0, 00m20s elapsed]
aws_instance_myapp_server: Still destroying... [id=i-095776ece185713ae, 00m30s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0ccc1c18b3a01e71b0, 00m30s elapsed]
aws_instance_myapp_server: Still destroying... [id=i-095776ece185713ae, 00m40s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0ccc1c18b3a01e71b0, 00m40s elapsed]
aws_internet_gateway.myapp_igw: Destruction complete after 48s
aws_instance_myapp_server: Still destroying... [id=i-095776ece185713ae, 00m50s elapsed]
aws_instance_myapp_server: Still destroying... [id=i-095776ece185713ae, 01m00s elapsed]
aws_instance_myapp_server: Destruction complete after 1m1s
aws_subnet.myapp_subnet_1: Destroying... [id=subnet-07bbc7d18a3245a8f]
aws_default_security_group.myapp_sg: Destroying... [id=sg-0d75bf1863327de55]
aws_default_security_group.myapp_sg: Destruction complete after 0s
aws_subnet.myapp_subnet_1: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-04dabca6095a83798]
aws_vpc.myapp_vpc: Destruction complete after 0s

Destroy complete! Resources: 6 destroyed.
```

- cleanup_state_files.png

```
@toobashafique065 /workspaces/Lab11 (main) $ cat terraform.tfstate
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 45,
  "lineage": "c1d66ff4-1548-5719-54a5-afb90a6ec625",
  "outputs": {},
  "resources": [],
  "check_results": null
}
@toobashafique065 /workspaces/Lab11 (main) $ cat terraform.tfstate.backup
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 38,
  "lineage": "c1d66ff4-1548-5719-54a5-afb90a6ec625",
  "outputs": {
    "aws_instance_public_ip": {
      "value": "3.29.15.224",
      "type": "string"
    }
  },
  "resources": [
    {
      "mode": "managed",
      "type": "aws_default_route_table",
      "name": "main_rt",
      "provider": "provider[\"registry.terraform.io/hashicorp/aws\"]",
      "instances": [
        {
          "schema_version": 0,
          "attributes": {
            "arn": "arn:aws:ec2:ms-central-1:535942231766:route-table/rtb-0fa00af3be972e7d0",
            "default_route_table_id": "rtb-0fa00af3be972e7d0",
            "id": "rtb-0fa00af3be972e7d0"
          }
        }
      ]
    }
  ]
}
```

- cleanup_verify_no_secrets.png

```
@toobashafique065 /workspaces/Lab11 (main) $ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    .gitignore
    .terraform.lock.hcl
    aws/
    awscliv2.zip
    locals.tf
    main.tf

nothing added to commit but untracked files present (use "git add" to track)
@toobashafique065 /workspaces/Lab11 (main) $ cat .gitignore
.terraform/*
*.tfstate
*.tfstate.*
*.tfvars
*.pem
```